Kalman Filter Tuning and Observability Analysis

Description

This project extends a linear Kalman Filter implementation by evaluating estimator consistency and system observability. In addition to state estimation, three performance metrics are studied: the mean squared error (MSE), the average normalized error estimate squared (ANEES), and the average normalized innovation squared (ANIS). The project also examines how observability changes under modifications to the measurement and dynamics models.

System Model

The system is the same linear position-velocity model used in the previous Kalman Filter project. The state is

$$x = \begin{bmatrix} p_x \\ p_y \\ v_x \\ v_y \end{bmatrix}$$

with linear dynamics and linear measurements. Process and measurement noise are modeled as zero-mean Gaussian random variables with covariances Q and R.

Truth data are generated using fixed baseline values of Q and R, while the estimator is run using scaled versions of these matrices to study the effect of covariance mismatch.

Consistency Metrics

Three metrics are used to evaluate estimator performance.

Mean Squared Error (MSE):

$$\mathrm{MSE} = \frac{1}{K} \sum_{k=1}^{K} \underline{e}_k^\top \underline{e}_k$$

where

$$\underline{e}_k = \underline{x}_k - \hat{\underline{x}}_k$$

is the state estimation error.

Average Normalized Error Estimate Squared (ANEES):

$$\mathrm{ANEES} = \frac{1}{KN} \sum_{k=1}^{K} \underline{e}_k^\top \hat{\mathbf{P}}^{-1} \underline{e}_k$$

This metric computes the actual estimation error to the filter's predicted covariance. For a consistent estimator, the ANEES should approach 1.

Average Normalized Innovation Squared (ANIS):

$$\mathrm{ANIS} = \frac{1}{KM} \sum_{k=1}^{K} \underline{r}^\top \hat{\mathbf{S}}^{-1} \underline{r}$$

where

$$\underline{r} = \underline{z} - \mathbf{H}\hat{\underline{x}}$$

is the innovation and

$$\mathbf{S} = \mathbf{H}\hat{\mathbf{P}}\mathbf{H}^\top + \mathbf{R}$$

is its covariance. Like the ANEES, the ANIS should also approach 1 for a well-tuned estimator.

Covariance Scaling Study

The truth data and measurements are generated using fixed baseline covariance matrices Q and R. The estimator is then run over 49 combinations of scaled covariance matrices using the factors

$$\left\{ \frac{1}{16}, \frac{1}{9}, \frac{1}{4}, 1, 4, 9, 16 \right\}$$

The process noise covariance scaling increases down the rows of the analysis matrices, and the measurement noise covariance scaling increases across the columns. For each combination, the MSE, ANEES, and ANIS metrics are computed and stored in 7×7 matrices.

This study shows how sensitive the estimator is to incorrect assumptions about the process and measurement uncertainty.

Observability Analysis

The observability of the system is analzyed using the observability matrix

$$O = \begin{bmatrix} \mathbf{H} \\ \mathbf{HF} \\ \vdots \\ \mathbf{HF}^{l-1} \end{bmatrix} \qquad l = \left\lceil \frac{N}{M} \right\rceil$$

The singular value decomposition (SVD) of the observability matrix is then computed:

$$O = USV^\top$$

The singular values indicate the strength of observability in different directions of the state space, while the rows of $V^\top$ give the corresponding observability vectors. Zero or near-zero singular values indicate non-observable or weakly observable directions.

Modified Systems

Two system modifications are studied.

Measurement Modification

The measurement model is changed so that one position-related measurement is replaced by a velocity measurement component. The observability matrix, singular values, and observability vectors are then recomputed. The modified H matrix is

$$\mathbf{H} = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Dynamics Modification

Keeping the modified measurement model, the dynamics are further changed so that the state coordinates rotate at a constant rate. This alters the coupling between the states and changes the observability of the system. The resulting observability matrix and singular values demonstrate that both the dynamics model and the measurement model joint determine which state directions are observable. The modified Φ matrix is

$$\Phi = \begin{bmatrix} 0 & \gamma & 1 & 0 \\ -\gamma & 0 & 0 & 1 \\ 0 & 0 & 0 & \gamma \\ 0 & 0 & -\gamma & 0 \end{bmatrix}$$

where $\gamma = \frac{4\pi}{180}$ corresponds to a rotation of the coordinate sytem of the state space at the rate of 4 degrees per second clockwise.

Results

Figures 1, 2, and 3 show the MSE, ANEES, and ANIS values across all 49 covariance-scaling combinations.

0.17	0.16	0.15	0.2	0.41	0.67	0.96
0.18	0.16	0.14	0.17	0.29	0.47	0.67
0.2	0.17	0.15	0.14	0.2	0.29	0.4
0.28	0.23	0.18	0.14	0.13	0.16	0.19
0.43	0.34	0.25	0.17	0.13	0.13	0.13
0.57	0.45	0.33	0.21	0.14	0.13	0.12
0.69	0.55	0.41	0.24	0.16	0.14	0.13

Figure 1: MSE under process and measurement covariance scaling

16.66	13.13	10.12	7.54	6.42	6.05	5.83
12.56	9.37	6.79	4.7	3.82	3.55	3.41
9.3	6.39	4.16	2.53	1.89	1.7	1.61
7.0	4.3	2.32	1.04	0.63	0.52	0.47
6.36	3.66	1.75	0.58	0.26	0.19	0.16
6.37	3.56	1.62	0.48	0.18	0.11	0.09
6.5	3.58	1.59	0.44	0.14	0.09	0.06

Figure 2: ANEES under process and measurement covariance scaling

16.09	9.22	4.23	1.16	0.35	0.19	0.13
15.8	9.05	4.13	1.11	0.32	0.17	0.11
15.34	8.82	4.02	1.06	0.29	0.14	0.09
14.19	8.29	3.84	1.01	0.26	0.12	0.07
12.17	7.39	3.55	0.96	0.25	0.11	0.07
10.48	6.59	3.28	0.92	0.25	0.11	0.06
9.1	5.9	3.04	0.89	0.24	0.11	0.06

Figure 3: ANIS under process and measurement covariance scaling

Figures 4–6 show the observability matrices and singular values for the modified and unmodified systems.

$$s = \{1, 1, 1, 1\}$$
$$\boldsymbol{V}^\top = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Figure 4: Observability matrix and singular values for unmodified system

$$s = \{1.31, 1, 0.54, 0\}$$
$$\boldsymbol{V}^\top = \begin{bmatrix} 0 & 0 & 0.92 & 0.38 \\ 0.71 & 0.71 & 0 & 0 \\ 0 & 0 & -0.38 & 0.92 \\ -0.71 & 0.71 & 0 & 0 \end{bmatrix}$$

Figure 5: Observability matrix and singular values for measurement-only modification

$$s = \{1.31, 1, 0.55, 0.01\}$$
$$\boldsymbol{V}^\top = \begin{bmatrix} 0.03 & -0.03 & -0.92 & -0.38 \\ 0.71 & 0.71 & 0 & 0 \\ 0.06 & -0.06 & 0.39 & -0.92 \\ 0.7 & -0.7 & 0 & 0.1 \end{bmatrix}$$

Figure 6: Observability matrix and singular values for measurement and dynamics modification

Observations

MSE

The highest MSE values were the ones with Q and R having opposing end scale factors. For example Q scaled by 1/16 and R scaled by 16 leads to a large MSE because our filter will almost solely follow the dynamics. Similarly, Q scaled by 16 and R scaled by 1/16 will have the opposite effect on the filter, essentially following only the measurements. Both of these cases and cases where Q and R are scaled similarly apart produce large MSE's because our dynamics and measurements are noisy by themselves. It then follows logically that when Q and R are scaled the same amount, it leads to the lowest (and best) MSEs.

One interesting thing of note is that our MSEs tend to be slightly better, in this scenario, when Q and R are scaled the highest (but still the same amount). This is actually a minor bug in our setup, which is explained in two ways. The first is that the covariance matrix P evolves according to the discrete-time Riccati recursion. The solution to this equation requires an infinite time horizon. So, if we increased the number of steps in our simulation, our MSE would be equal along the big diagonal. The second explanation is that when we scale Q by some value k, we should also scale the prior P0 by the same amount. Scaling P0 by the same amount as Q fixed this behavior along the diagonal, as expected.

ANEES

The highest ANEES values are in the top left corner, where both Q and R are scaled to be very low. This leads to a very small covariance matrix which harshly penalizes our larger errors. A high ANEES score means our filter is overconfident, which makes sense with Q and R values lower than truth. Similarly, the lowest ANEES values are in the lower right corner, where both Q and R are scaled to be very high. This leads to a large covariance matrix which forgives even the largest errors. A low ANEES score means our filter is underconfident, which makes sense with Q and R values higher than truth. Lastly, our closest values to 1 are in the middle, where the filter Q and R most accurately represent the truth Q and R.

ANIS

Similar to ANEES, the ANIS values are highest in the top left corner, and the lowest in the bottom right corner. The main difference with ANIS is that the score is more sensitive to changes in R. This is because ANIS directly deals with R when it calculates difference from expected measurement. The fact that the R noise is higher than the Q noise in generation and in the baseline estimation is also a factor. We again notice that our values closest to 1 are at the positions closest to the truth Q and R, with R being more sensitive here too.

Observability

For both the unmodified case and the case where the measurements and dynamics were modified, we notice that all states in the system are observable, though the unmodified case had "more" observable states.

In the case where only the measurements are modified, we note one linear combination of states that is "unobservable". The unobservable direction is $-p_x + p_y$ as shown by $s_4 = 0$ and row 4 of $V^T = -sin\theta$, $cos\theta$, 0, 0.

Discussion

The covariance scaling study shows that estimator performance depends strongly on the assumed values of Q and R. When these covariances are poorly tuned, the filter may still produce estimates, but the resulting uncertainty no longer matches the true estimation error. This is reflected by ANEES and ANIS values that deviate from 1.

The observability analysis shows that accurate estimation is not determined by the measurement model alone. The dynamics also influence which directions in state space can be reconstructed from the available measurements. Modifying the measurement model can reduce observability, while modifying the dynamics can restore coupling that improves it.

Together, these results highlight two central ideas in state estimation: a filter must be both well tuned and applied to a system whose states are sufficiently observable.